

Information, Entropy, and Semantic Stability: Abstraction Beyond Compression

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Abstract

Information-theoretic accounts of intelligence and cognition have historically centered on compression, prediction, and entropy minimization. While these notions successfully explain statistical efficiency and generalization, they do not by themselves account for semantic stability: the persistence of meaning under perturbation, recombination, and historical variation. This paper develops a structural account of abstraction as a constraint-sensitive process operating under entropy, rather than as an entropy-reducing operation itself. By distinguishing compression from semantic equivalence, and entropy from invariance, we show that abstraction corresponds to a phase transition in the organization of replayable histories. Entropy constrains abstraction, but does not define it. The resulting framework clarifies why compression-heavy systems may remain semantically brittle, and why abstraction exhibits threshold behavior in complex adaptive systems.

1 Introduction

Information theory has provided some of the most powerful tools for understanding learning, inference, and adaptation. From Shannon entropy to rate–distortion theory, compression has been treated as both an explanatory and normative principle: intelligent systems, it is often argued, are those that discover compact representations of the regularities in their environment [?, ?].

Yet compression alone does not explain why certain structures remain meaningful across transformations, interventions, and recombinations, while others collapse under minimal perturbation. Systems may achieve extraordinary compression while remaining semantically fragile, unable to preserve invariant structure when faced with novel contexts. This observation motivates a distinction that has remained underdeveloped in information-theoretic accounts: the distinction between reducing description length and stabilizing semantic equivalence.

This paper advances a structural account of abstraction that treats entropy as a constraint rather than as a defining principle. Abstraction is understood as the stabilization of equivalence classes over replayable construction histories, subject to entropic limits. Compression may facilitate this process, but abstraction itself corresponds to a different organizational regime—one characterized by robustness, invariance, and historical reconstructibility.

The central claim is not that entropy is irrelevant to abstraction, but that entropy operates orthogonally: it limits what can be stabilized, how deeply, and at what cost. Semantic stability

emerges when systems organize their histories in ways that survive entropic perturbation, yielding a qualitative transition in behavior.

2 Entropy as Constraint, Not Essence

Entropy enters cognitive and computational theories through multiple channels: uncertainty reduction, coding efficiency, thermodynamic cost, and irreversibility. These notions are often conflated, leading to ambiguity about what entropy is supposed to explain. In the present framework, entropy is treated neither as a semantic primitive nor as the substance of meaning. Instead, it functions as a boundary condition on what kinds of semantic structures can be maintained.

Shannon entropy quantifies uncertainty over symbol distributions, not over semantic effects. A representation may be maximally compressed in the Shannon sense while still encoding no stable semantic structure beyond the dataset on which it was trained. Conversely, a system may tolerate relatively high entropy while preserving invariant semantic relations across contexts.

Thermodynamic entropy further constrains abstraction by imposing costs on memory, replay, and revision. Replayable histories require energy to maintain, reconstruct, and perturb. These costs place upper bounds on the depth and breadth of abstraction, but they do not determine its form. Semantic equivalence is not identical to energetic efficiency, even though the two are coupled.

Proposition 2.1. *Entropy constrains the feasibility of abstraction without determining its semantic content.*

This distinction parallels that between syntax and semantics: entropy governs the space of possible encodings, while abstraction governs which equivalence relations remain stable within that space.

3 Compression and Semantic Degrees of Freedom

Compression reduces the number of degrees of freedom required to encode data. Abstraction reduces the number of semantic representatives required to explain effects. These operations are related but non-identical.

A compressed representation may discard information that is irrelevant for prediction on a fixed distribution, yet that same information may be crucial for preserving semantic identity under intervention. Compression optimizes for average-case efficiency; abstraction optimizes for invariance across histories.

Consider two constructions that yield identical outputs on all observed inputs but differ in how those outputs were generated. Compression collapses them; abstraction distinguishes them until their equivalence is stabilized under replay. This distinction explains why compressed systems may generalize well yet fail catastrophically under recombination or counterfactual manipulation.

Definition 3.1. *The semantic degrees of freedom of a system are the number of distinct construction histories that must be preserved to maintain invariant effects under replay.*

Abstraction reduces semantic degrees of freedom; compression reduces descriptive degrees of freedom. The former requires the latter only insofar as entropy budgets permit.

4 Replay, Entropy, and Irreversibility

Replayability introduces an asymmetry into system dynamics. Construction histories accumulate irreversibly, even when abstractions are later formed. This asymmetry mirrors thermodynamic irreversibility, but the correspondence is structural rather than literal.

Replay allows systems to distinguish essential from accidental structure by comparing alternative histories. Each such comparison incurs entropic cost: storing alternatives, reconstructing trajectories, and evaluating perturbations require resources. As a result, abstraction cannot proceed indefinitely. It must operate within an entropy budget that limits how many distinctions can be tested and stabilized.

Theorem 4.1. *There exists an upper bound on abstraction depth determined by the entropy budget available for replay.*

This bound explains why abstraction often exhibits punctuated dynamics rather than smooth accumulation. Systems accrue construction histories until replay becomes costly, at which point abstraction collapses multiple histories into a stable equivalence. The process then repeats at a higher level.

5 Phase Transitions in Semantic Organization

The transition from compression-dominated organization to abstraction-dominated organization exhibits features analogous to phase transitions in complex systems. Below a critical threshold of replay capacity, systems behave as pattern extractors: they compress, predict, and interpolate. Above that threshold, they reorganize around invariant semantic effects.

This transition is not merely quantitative. It involves a qualitative change in what counts as structure. Prior to abstraction, structure is statistical; afterward, it is historical and counterfactual.

Definition 5.1. *A semantic phase transition occurs when invariant equivalence classes emerge that are stable under admissible perturbations of construction history.*

Such transitions depend on control parameters including memory depth, replay fidelity, and entropy export capacity. When these parameters cross critical values, semantic basins form: regions of construction space that map to the same semantic effect.

These basins may deepen, merge, or split as abstraction evolves, but their existence marks a categorical shift in system behavior.

6 Stability, Robustness, and Entropic Noise

Semantic stability is not absolute. It is always defined relative to a class of perturbations and a noise model. Entropy introduces variability at every level: stochasticity in construction, noise in

replay, and drift in representations.

Abstraction succeeds when semantic equivalence remains invariant across this variability. Robust abstraction therefore requires not minimal entropy, but controlled entropy: enough variability to test invariance, but not so much that replay collapses.

This perspective clarifies why systems optimized purely for entropy minimization may fail to abstract. Excessive regularization suppresses the very variability needed to identify invariant structure.

Proposition 6.1. *There exists an optimal entropy regime for abstraction, lying strictly between maximal order and maximal disorder.*

This result aligns with observations from complex adaptive systems, where maximal adaptability occurs near criticality rather than at extremes.

7 Relation to Information-Theoretic Frameworks

The framework developed here is compatible with, but not reducible to, established information-theoretic approaches. Minimum description length captures trade-offs between model complexity and fit, but does not address semantic invariance under replay. Information bottleneck methods formalize relevant information compression, but presuppose a task-defined notion of relevance.

Free-energy-based accounts emphasize prediction error minimization, yet prediction accuracy does not entail abstraction. Semantic stability requires preserving equivalence across unobserved interventions, not merely minimizing surprise.

These frameworks can be understood as operating within a single semantic basin. Abstraction governs the formation and stabilization of those basins themselves.

8 Implications for Scientific Explanation

Scientific theories exemplify abstraction beyond compression. A successful theory does not merely compress observations; it identifies invariant structures that remain stable across experimental interventions, measurement regimes, and historical reinterpretations.

From this perspective, scientific progress involves the stabilization of increasingly abstract equivalence classes over empirical histories. Entropy constrains how finely these classes can be resolved, but does not define their content.

This view explains both the power and the fragility of scientific abstraction: theories persist when their invariances survive perturbation, and collapse when new evidence reveals instability in their equivalence classes.

9 Conclusion

Entropy constrains abstraction without defining it. Compression facilitates abstraction without guaranteeing it. Semantic stability emerges when replayable histories are organized into invariant equivalence classes that survive entropic perturbation.

By distinguishing entropy from semantics, and compression from abstraction, this paper clarifies why intelligent organization exhibits threshold behavior and why purely compression-driven systems remain brittle. Abstraction is not the elimination of information, but the stabilization of meaning under constraint.